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SUTRAM 2026

IDEA SUBMISSION

AI-POWERED CYBER INVESTIGATION & DIGITAL FORENSICS HACKATHON

Problem Statement ID	—
Theme	**Open Innovation**
Team Name (registered on portal)	**Merge Conflicts**

02 TEAM DETAILS

FORENSIQ · 4 members

01 TEAM LEADER

Pratibha

CSE · 2nd Year

02 MEMBER

Namita

CSE · 2nd Year

03 MEMBER

Muskan

CSE · 2nd Year

04 MEMBER

Pavani

CSE · 2nd Year

IDEA TITLE

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THE PROBLEM

Attackers are multi-vector. Your tools aren't.

3.4B	**Phishing emails** Sent daily worldwide
$10B	**Vishing losses** Annual cost of voice scams
78%	**Multi-channel attacks** Use 2+ attack vectors

No existing tool correlates evidence across email, phone, website and documents into one forensic case.

OUR SOLUTION

FORENSIQ — Unified Multi-Vector Digital Forensics

One platform. Four channels. One forensic truth.

- AI/ML-driven email, phone, website and document forensics in one case file
- Ingest evidence from email, SMS, chat, image or document
- Phone-number check to tell whether a call is genuine
- One 0–100 risk score and a correlation graph across all channels
- Blockchain-backed chain-of-custody for every piece of evidence
- Exportable PDF forensic report for investigators

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TECHNICAL APPROACH

STACK & TECHNOLOGIES

- Python · Flask · Docker
- scikit-learn (TF-IDF + Naive Bayes)
- SQLAlchemy / SQLite · ReportLab PDF
- web3.py blockchain ledger
- TinyFish live web agent

~97% holdout accuracy
on ~15.3K real emails (Enron, SpamAssassin, Nazario)

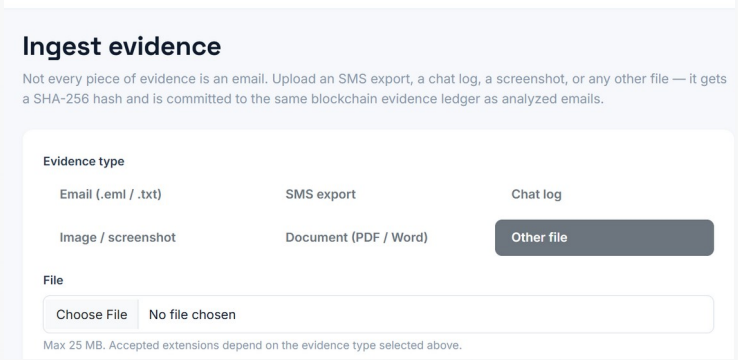
PIPELINE: INPUT → AI/ML → ENRICHMENT → OUTPUT

1 INPUT	Email · Phone numbers · URLs · Documents & images
2 AI/ML ANALYSIS	Phishing classifier · NLP social-engineering cues · Anomaly detection
3 ENRICHMENT	GeoIP, WHOIS, SPF/DKIM/DMARC · PhishTank · TinyFish live web checks
4 OUTPUT	Risk score (0–100) · Correlation graph · Blockchain ledger · PDF report

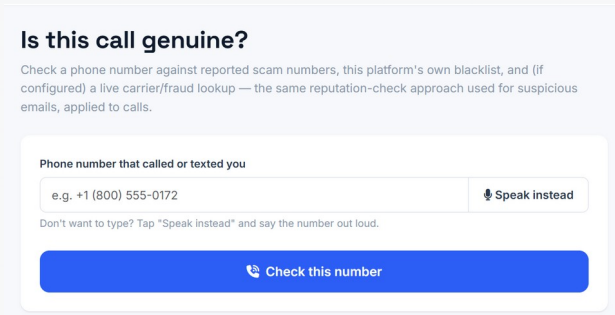
05 FEASIBILITY AND VIABILITY

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WORKING PROTOTYPE



Ingest evidence: email, SMS, chat, image, document



Is this call genuine? Phone-number check

ROADMAP (FUTURE SCOPE)

01 Browser & mobile app
real-time warnings on suspicious emails, links and calls

02 Voice & deepfake checks
analyse call audio and AI-made voices or images

03 SIEM integrations
push unified cases into SOC and cyber-crime workflows

04 Public-chain anchoring
anchor evidence hashes on a public blockchain

05 Multilingual detection
Hindi, Hinglish and regional-language scams

06 Threat-intel sharing
share anonymised attacker indicators across organisations

Built: email, phone, website & document forensics, evidence ledger, correlation graph, live checks.

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06 IMPACT AND BENEFITS

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TARGET USERS & IMPACT

Who benefits: SOC & incident-response teams, banks & platforms, cyber-cells and individuals.

4 → 1
channels, one case file

0–100
unified risk score

~97%
holdout accuracy, ~15.3K emails

$10B
voice-scam losses targeted

WHY FORENSIQ WINS

Capability	Email gateways	Caller-ID apps	URL & file scanners	FORENSIQ
Email phishing & spoofing	✓	—	—	✓
Phone-number scam reputation	—	✓	—	✓
Live link & website checks	Partial	—	✓	✓
Document & image tamper forensics	—	—	—	✓
One case across every channel	—	—	—	✓
Blockchain chain-of-custody	—	—	—	✓

Other tools detect on one channel. FORENSIQ investigates across all of them and proves the evidence was not tampered with.

Revenue: SaaS for SOC teams · pay-per-use API · cyber-cell licences · free tier

Compared by tool category, based on typical capabilities; not a feature audit of any single product.

RESEARCH AND REFERENCES

REFERENCES

Tech stack

Python + Flask · scikit-learn · SQLAlchemy / SQLite · ReportLab · web3.py + Docker

Services & threat feeds

TinyFish live web agent · PhishTank · GeoIP and WHOIS lookups · SPF / DKIM / DMARC checks

Datasets

Enron · SpamAssassin · Nazario corpora · ~15.3K real emails · ~97% holdout accuracy

RESEARCH & SOURCES

Market size & demand

Fortune Business Insights — Email Security Market, 2026–2034

fortunebusinessinsights.com

Competitor tools

Proofpoint vs. Mimecast comparison

proofpoint.com

Phishing threat data

FBI IC3 — 2025 Annual Report; APWG — Q1 2026 Trends Report

apwg.org

GitHub: github.com/Pratibha-igdtuw/email_protection_model

Demo video: drive.google.com/file/d/1XK5jYzewEkNSVu_MaUG66z5vsNQq_zvY/view?usp=drivesdk